

Course Outline

1. COURSE INFORMATION

Session Offered	Winter 2024	
Course Name	Ethics, Equity and Law in Engineering	
Course Code	ENGINEER 4A03	
Date(s) and Time(s) of lectures	C01: Wednesdays 7 – 10 PM (Virtual)	
Program Name		
Calendar Description	General introduction to the ethical, equitable and legal aspects of engineering practice. Topics include the engineering act and code of ethics, employment equity, legal duties and liabilities of the professional engineer, contracts, and the tort of negligence.	
Instructor(s)	Martha Kafuko, P.Eng.	E-Mail: kafukom@mcmaster.ca

2. COURSE SPECIFICS

Course Description	This course will explore a few themes and issues related to the question: What is a professional engineer's social responsibility? It is hoped that each of you will be able: • To understand the scope and breadth of your social/ethical responsibilities. • To translate a general awareness of ethical responsibility into specific engineering practices. • To contribute to the profession's activity in technology policy formation after you graduate and become a Professional Engineer.
Course Specific Policies	This course will be held online with synchronous and asynchronous components. Students must login to the lectures using their email addresses provided by McMaster (@mcmaster.ca). The lectures will be recorded and made available on Avenue to Learn each week. Late Work: Late submissions <u>will not be accepted</u> and the submitted materials will not be graded. Missed Work: If a student submits an MSAF the marks assigned to the missed work will be redistributed to other elements in the assessment category (i.e., quizzes). A group component cannot be redistributed using an MSAF. Educational Materials: The educational materials developed for this course, including, but not limited to, lecture notes and slides, handout materials, examinations and assignments, and any materials posted to Avenue, are the intellectual property of the course instructor. These materials have been developed for student use only and they are not intended for wider dissemination and/or communication outside of a given course. Posting or providing unauthorized audio, video, or textual material of lecture content to third-party websites violates the instructor's intellectual property rights, and the

	Canadian Copyright Act. Failure to follow these instructions may be in contravention of the university's Code of Student Conduct and/or Code of Academic Conduct and will result in appropriate penalties. Participation in this course constitutes an agreement by all parties to abide by the relevant University Policies, and to respect the intellectual property of others during and after their association with McMaster University and Mohawk College.	
3. SUBTOPIC(S)		
Week 1 Jan 8 - 12	Introduction, inquiry, history of engineer's concerns	
Week 2 Jan 15 - 19	Equity I	
Week 3 Jan 22 - 26	Risk & Technology Assessment, types of business's	Assignment 1: Robotic Limbs Due in Week 4
Week 4 Jan 29 - Feb 2	Ethics and Whistleblowing	Group Project: Stage One – Literature Review & Central Question Due in Week 5
Week 5 Feb 5 - 9	Micro/Macro Ethics & Central question	Assignment 2: Equity in Stream Restoration Due Week 6
Week 6 Feb 12 - 16	Ethics and Misconduct (mention ethics tribunal)	Group Project: Stage Two – Stakeholder Analysis Due in Week 9
Week 7	Midterm Recess: February 19 - 23	
Week 8 Feb 26 - Mar 1	Engineering Act	
Week 9 Mar 4 - 8	Engineering Law Introduction	Assignment 3: Liability for Robots Due in Week 10
Week 10 Mar 11 - 15	Tort Law	
Week 11 Mar 18 - 22	Contract Law	Group Project: Stage Three – Inquiry Report Due in Week 12
Week 12 Mar 25 - 29	Equity II	
Week 13 Apr 1 - 5	Any other topic	

Week 14 Apr 8-10	No Class
Classes end: Wednesday, April 10 th , 2024	
Note that this structure represents a plan and is subject to adjustment term by term. The instructor and the University reserve the right to modify elements of the course during the term. The University may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. <i>This lecture schedule is based upon current university and public health guidelines and may be subject to changes during the term. Any changes to the schedule or course delivery will be communicated on the course announcements section on Avenue to Learn. Please check the announcements prior to attending class.</i>	
4. ASSESSMENT OF LEARNING *including dates*	Weight
Quizzes*	30%
Assignments	25%
Stage One: Literature Review and Central Question	5%
Stage Two: Stakeholder Analysis	5%
Stage Three: Inquiry Report	30%
Participation	5%
TOTAL	100%
Percentage grades will be converted to letter grades and grade points per the University calendar. * Quizzes are estimated to take 5 minutes to complete; however, they are available for up to 30 minutes.	
5. LEARNING OUTCOMES	
1. Identify and analyze the essential characteristics of a complex problem from a sustainability perspective, including its ethical dimensions, risks, and uncertainties. (GA 4.5)	
2. Feel and explain a sense of respect for diversity, the environment, and past, present, and future generations in all engineering decisions. (GA 9.1, 9.2 and 9.3)	
3. Identify professional ethical dilemmas and competing stakeholder interests and develop conscientious, well-reasoned, professional responses. (GA 10.1, 10.2, 10.3)	
4. Identify and quantify short and long-term impacts of engineering on a scale ranging from the local to the global. (GA 9)	
5. Communication skills for effective teamwork, influence, and effectiveness. (GA 10)	
6. Thoughtfully consider personal and professional choices and career contributions. (GA's 8, 9, 10, 11)	
7. Be able to critically evaluate and apply knowledge, methods and skills procured through self-directed and self-identified sources and is aware of engineering societies and literature. (GA 12)	
6. COURSE OUTLINE – APPROVED ADVISORY STATEMENTS	
ANTI-DISCRIMINATION	
The Faculty of Engineering is concerned with ensuring an environment that is free of all discrimination. If there is a problem, individuals are reminded that they should contact the Department Chair, the Sexual Harassment Officer or the Human Rights Consultant, as soon as possible. http://www.mcmaster.ca/policy/General/HR/Discrimination Harassment Sexual Harassment-Prevention&Response.pdf	

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. It is your responsibility to understand what constitutes academic dishonesty.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at <https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/>

The following illustrates only three forms of academic dishonesty: The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g., the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

AUTHENTICITY / PLAGIARISM DETECTION

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g., A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com, please go to www.mcmaster.ca/academicintegrity.

COURSES WITH AN ON-LINE ELEMENT

Some courses may use on-line elements (e.g., e-mail, Avenue to Learn (A2L), LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

ONLINE PROCTORING

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

COMMUNICATIONS

It is the student's responsibility to:

- Maintain current contact information with the University, including address, phone numbers, and emergency contact information.
- Use the University provided e-mail address or maintain a valid forwarding e-mail address.

- Regularly check the official University communications channels. Official University communications are considered received if sent by postal mail, by fax, or by e-mail to the student's designated primary e-mail account via their @mcmaster.ca alias.
- Accept that forwarded e-mails may be lost and that e-mail is considered received if sent via the student's @mcmaster.ca alias.
- Check the McMaster/Avenue email and course websites on a regular basis during the term.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g., use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult [McMaster University's Academic Accommodation of Students with Disabilities policy](#).

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

McMaster Student Absence Form (MSAF): In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar "Requests for Relief for Missed Academic Term Work".

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office normally within 10 working days of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests. <http://www.mcmaster.ca/policy/Students-AcademicStudies/Studentcode.pdf>

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, including lectures by University instructors. The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.